

Daniel Johann Thomas

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EDUCATION

St. Stephen's College, University of Delhi

Expected 2027

B.A. (Honours) Economics, CGPA 9.39/10 (through Semester IV; 10-point scale: O = 10, A+ = 9, A = 8)

- **Professor Nagpaul Research Fellow (2025):** the only second-year Economics student selected by the Mathematics Department, St. Stephen's College, for game theory with incomplete information under Prof. Anant Singh.
- *Mathematics:* Real Analysis (O), Linear Algebra (O), Graph Theory (O), Adv. Mathematical Methods (A+), Calculus (A+)
- *Statistics & Econometrics:* Statistics with R (O), Intermediate Statistics (A+), Introductory Econometrics (A+)
- *Economics:* Intermediate Microeconomics I–II (A+), Intermediate Macroeconomics I–II (A+)
- *In progress:* Advanced Econometrics, Game Theory, Growth, Development, Financial Modelling

RESEARCH

Priced In, Bit by Bit: Measuring Information Aggregation in Prediction Markets [Paper]

Jun–Sep 2026

- Built a 43-million-row hourly panel from Polymarket's API (50,000 markets); cut memory by **86.9%** (5.35 GB → 699 MB).
- Applied **Shannon's information theory** to quantify how much outcome uncertainty prices resolve: 38% on average, and up to 85% one day before close in long-lived markets, which learn about six times faster near resolution.
- Tested price predictability via **Samuelson's martingale property**; deviations from a fair game are statistically significant but economically negligible ($R^2 = 0.021$), as a one-cent move is followed by an average reversal of only **0.17 cents**.

Drawing the Last Straw: Deception and Competition in a Duopoly (Nagpaul paper) [Paper]

Nov 2025–May 2026

- Modeled duopolists competing in deception against private tolerance thresholds; proved no **pure strategy equilibrium**.
- Derived the **mixed strategy equilibrium** from indifference conditions and verified it numerically in Python; firms leapfrog each other, deceiving **41% more than a monopolist** while each earns under half the monopoly profit.
- Extended the model to **incomplete information on consumer tolerance**; uncertainty makes firms cautious, but once a tolerant market seems likely they gamble, making **market collapse** an equilibrium outcome rather than a mistake.

Income Segregation in an Agent-Based Housing Market [GitHub]

Aug 2025–Jan 2026

- Extended **Schelling's segregation model** with a budget-constrained rental market; agents have **Cobb–Douglas preferences** over neighborhood income mix and consumption, calibrated to Delhi income microdata (CMIE).
- Ran sealed-bid auctions for **1M agents** across 100 neighborhoods per round, with endogenous rents and displacement of priced-out tenants; tracked sorting via a **Theil decomposition** into within- and between-neighborhood inequality.
- Against a no-policy Monte Carlo baseline, a **below-market set-aside** (20% of units for the bottom 30% of earners) cut the Theil index by **16%** but left **48%** of reserved units vacant and raised the unhoused share by 9.1 pp.

State Data Center Policy Competitiveness Index (with Nathanya Eby Thomas) [Paper]

Aug 2026

- Ran a **Monte Carlo sensitivity analysis** of a 17-state, 33-incentive index built from state policy documents, re-ranking all states under **flat Dirichlet** weights sampled uniformly over the simplex of its 5 dimensions, not one weight at a time.
- Used **swing-factor analysis**, correlating simulated ranks with each dimension's weight, to find what drives each state.
- Found balanced Rajasthan was the only state **never below 4th**, while Chhattisgarh ranged from **1st to 16th** with 35/60 on fiscal incentives but 0/12 on infrastructure; up-weighting **assured power and water** lifted the most states (8 of 17).

Profits, Prices, and the Power of Information (with Madhura Mukund, Tanishi Singhal) [Article]

Aug 2025

- Published in *Vitta* (SRCC, Delhi), **India's oldest undergraduate finance journal**; **10 of 1,700** entries selected.
- Used price theory and the **Grossman–Stiglitz paradox** to analyze the competitive race for information in algorithmic markets, framing profit as the opportunity cost of secrecy; some mispricing must persist.

TECHNICAL SKILLS

Programming: Python (NumPy, Pandas, SciPy, SymPy, Numba, Matplotlib), R (tidyverse, ggplot2), Git, L^AT_EX, Excel

Computational economics: Dynamic programming, Markov chains, Monte Carlo methods ([QuantEcon](#))

HONORS & AWARDS

- **Project Euler:** top 5.6% of 925K users globally; solved 58 mathematical coding problems (number theory, algorithms).
- **Lyons Range 15.0 (2026):** National 2nd Runner-Up, mock stock competition, Kirori Mal College, University of Delhi.
- **Toastmasters (2023):** 1st in Table Topics (extempore) and 3rd in Evaluation Speech at the Annual Club Contest.

LEADERSHIP & SERVICE

- **Junior Strategist, Indian Game Theory Society, St. Stephen's Chapter (2025–26):** designed the event roadmap; organized lectures for 40+ students; designed live games and wrote articles teaching game theory.
- **Editorial Volunteer, Economics Society (2024–25):** editorial team, annual journal *Towards Equilibrium*.